1. Analysis and Programming

Def 1.1 (Limit). *A limit is the value that a function approaches as the input approaches some value.* $\lim_{x \to c} f(x) = L$.

Thm 1.2 (Taylor Expansion). $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$.

Lem 1.3 (Squeeze Theorem). *If* $g(x) \leq f(x) \leq h(x)$ *and* $\lim_{x \to c} g(x) = \lim_{x \to c} h(x) = L$, *then* $\lim_{x \to c} f(x) = L$.

Rem 1.4. *Limit laws apply only when the limits exist.*

Imp 1.5. *The Big-O notation describes the limiting behavior of a function.*

Ex 1.6. $O(n \log n)$ *is the complexity of Merge Sort.*

Code Ex 1.7

```
// Merge Sort Example
void mergeSort(int arr[], int l, int r);
```

Algo 1.8

```
Target x in sorted arr[]
1. l=0, r=n-1
2. while l <= r:
   m = (l+r)/2
   if arr[m] == x: return m
   if arr[m] < x: l = m+1
   else: r = m-1
```

Listing 1. Binary Search

1.1. Complexity Table

Algorithm	Complexity
Quick Sort	$O(n^2)$ / $O(n \log n)$
Binary Search	$O(\log n)$
DFS / BFS	$O(V+E)$